A Sequential Decomposition Framework and Method for Calibrating Dynamic Origin-Destination Demand in a congested network

Xianbiao Hu*, Yi-Chang Chiu, Jorge A. Villalobos, Eric Nava

Abstract— **This paper presents a two-stage model to calibrate the time-dependent, dynamic Origin-Destination (O-D) demand under congested traffic conditions. The first-stage model estimates O-D trip rates by minimizing link demand deviation with a one-norm formulation approach, so that over the calibration time period, the traffic demand on calibration links matches with the link demand from the field data. Due to its linear model structure, first-stage model is more computationally effective and solvable on large real-life networks compared with the commonly seen least-square formulation. Then, a time-dependent user equilibrium traffic assignment (TDUETA) model is formulated at the second-stage to adjust the departure time profile iteratively, aiming to match the calibrated result with the field observed dynamic traffic condition, i.e. time dependent speed profile. The second-stage model starts from the concept of demand-capacity-volume relationship at a congested road segment where demand exceeds supply, utilizes shockwave theory to capture the differences between true demand and volume output, together with the idea of using travel time propagation between origin and bottleneck locations to infer real demand at origin location. The two-stage model was implemented and tested in a case study in Tucson, AZ as an experimental proof of concept, which demonstrated the effectiveness of the proposed calibration framework and method under circumstances in which departure time profile was systematically distorted and initial demand solutions deviated from the true O-D matrices.**

Index Terms— **O-D Calibration; Simulation Model Calibration; Dynamic Traffic Assignment; One-norm formulation; Shockwave propagation;**

I. INTRODUCTION

Calibrating a Dynamic traffic assignment (DTA) model is a necessary yet challenging step, where a baseline model needs to be firstly established and then calibrated to ensure it properly matches with the time-dependent traffic patterns observed from the field data. In particular, adjustment of origin-destination (O-D) flows, i.e. O-D demand calibration to improve simulation realism is one of the most fundamental important topics particularly for a large size network under congested traffic conditions where the efficiency and accuracy of model calibration remain challenging.

A two-stage model is built to further research in this field, with the first-stage problem being to estimate O-D trip rates by minimizing link demand deviation with a one-norm formulation approach, aiming to efficiently match the traffic demand on calibration links with the link demand from the field data. Then, the second-stage problem is formulated as a time-dependent, user equilibrium traffic assignment model to adjust departure time profile iteratively, aiming to match the calibrated result with field observed dynamic traffic condition, i.e. time dependent traffic speed profile.

The proposed sequential decomposition framework departs from previous related literature mainly in two aspects. First, compared with the commonly seen least-square type of bi-level model formulation, the one-norm formulation proposed in this paper is more computationally effective and solvable on large real-life networks due to its linear model structure. Second, the second-stage model adopts an innovative approach to utilize field data in the calibration process, specifically, starts from the concept of demand-capacity-volume relationship at a congested road segment where demand exceeds supply, utilizes shockwave theory to capture the differences between true demand and volume output. In other words, by incorporating time dependent speed profile along with link volume data, derives an estimate of "arrival" or "demand" information for the congested bottleneck locations that need calibration. Then, based on the idea of using travel time propagation between origin and bottleneck locations, the real traffic demand at origin locations can be inferred through proper spatial-temporal mapping. Consequently, the O-D calibration results can properly depict the traffic demand pattern at the bottleneck location.

It's worth noting that "demand" and "count" are two related but different words used in this research. Here, link demand, which is the calibration target of first-stage model, refers to the total number of vehicles that passes a given location within certain time period (say 24 hours) which includes both congested hours and non-congested hours, under which condition the vehicles in queue will be discharged and able to pass the bottleneck location, i.e. the true demand can be captured, while 'count' means the number of vehicles passing through a link within a shorter time period. Once the total link demand is calibrated to within a reasonable range, which is the prerequisite of next stage calibration, second-stage model starts to adjust vehicle's departure time profile according to

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the methodology defined in this paper, so that link's time-dependent speed profile will also match that of ground truth. Time-dependent link count or volume profile calibration, although a natural outcome once the speed profile matches, is not a calibration target for this paper per se.

The two-stage model was implemented and tested in a case study in Tucson, Arizona as an experimental proof of concept. The testing results demonstrated the effectiveness of the proposed O-D calibration framework and method under situations in which departure time profile was systematically distorted and initial traffic demand solutions deviated from the true O-D matrices.

II. LITERATURE REVIEW

As mentioned in Section I, the first main difference this research brings is from the model formulation perspective. In prior studies, least-square bi-level type of two-norm formulation is a commonly seen approach in which usually the upper level problem is formulated to minimize the degree of deviations between calibrated demand and the observed O-D matrices, and lower level problem is formulated as linear or non-linear constraints or traffic assignment problem [1-3]. Variations of bi-level formulation approach can also be found in the literature, such as a modified bi-level optimization algorithm proposed by Verbas et al. [4] with techniques to reduce the number of variables and parameters; a model to estimate the mean and covariance of stochastic multi-class OD matrices by Shao et al. [5]; and a one-level convex optimization formulation by Shen and Wynter to approximate the bilevel structure [6]. In addition, simultaneous perturbation stochastic approximation (SPSA) algorithm, due to its advantages in allowing general formulations that can include a wide range of sensor data measurements, has been applied in prior literatures for O-D calibration purposes [7-9]. Due to the two-norm formulation, its computational efficiency and application in large-scale network remain to be challenging.

Another less seen formulation from the least-squares problem (or quadratic) is the one-norm formulation in which the objective function is the absolute value of the deviation. Generally speaking, one-norm formulation is more computationally effective and solvable on large real-life networks due to its linear model structure, which makes it solvable by most existing LP solvers. Sherali et al. [10] proposed a least absolute norm approach for estimating split parameters where two optimization methods were proposed to estimate the optimal values to be used in OD calibration process, but the model objective per se, isn't on the OD calibration. In addition, the model is based on the "entering and exiting traffic volumes through an intersection or a small freeway segment" which may have limitations in applications. The authors of this paper also formulated the O-D calibration problem in a one-norm approach[11], from which this research manuscript is derived.

The second main difference this research brings is from the calibration target and data perspective, where in the prior studies traffic counts are frequently used as the calibration target, examples can be found in Carey et al. [12], Cascetta et al. [13], Zhou and Mahmassani [14], Bierlaire and Toint [15], and so on. It is, however, a widely accepted understanding that count data is not the ideal calibration target. Learning from the traffic flow fundamental diagrams, one can see that a low volume may result from either a light or heavy congestion situation. Some later researches started to take in other data in the modeling process for calibration purposes. Ashok and Ben-Akiva [16] proposed a model that can match with field observed travel time fluctuations, and the research of using time-dependent speed profile as calibration target can also be found in [17, 18]. Besides commonly seen traffic counts, speed and density, other data such as Call Detail Records (CDR) [19], routing information for a fraction of vehicles on the network [20], automatic vehicle identification (AVI) counts data [3] are also used in the O-D estimation process. In these studies, data on the calibration links of interest are typically used as calibration target only, although through proper modeling, the true traffic demand information could be revealed based on the bottleneck volume throughput and density information.

In addition, in prior studies the split rates are usually modeled as fixed [12, 13, 15], or determined by DTA model [2, 16, 17], or from optimization algorithm [21-23]. It's worth noting that in this research, split rate is adjusted in the first-stage model, while the second-stage model fixes split rate unchanged and adjusts vehicle departure time profile instead. In terms of deployment environment, the calibration model developed in this model takes link-based speed, density and volume profile as input, and could be designed to take in real time traffic data feed, similar to [14, 24-26], and have the potential to be deployed in both offline and online fashion, as also discussed in [27-29]. The online deployment will actually benefit from the computational effective one-norm formulation design.

Table 1 below summarizes the O-D calibration research approaches in the representative past literature, and systematically compares their differences with the research approach proposed in this paper. The literature review above highlights the fact that model formulation wise, the O-D calibration models were mostly based on least-square type of two-norm formulations, which due to computational intractability, are ineffective for large-scale networks. In addition, field observed data was typically used only as calibration target, but not utilized in the demand calibration process although it could be very informative in telling the dynamic traffic condition and indicating the real traffic demand at bottleneck locations. Section III below presents a two-stage O-D calibration model in details to address these concerns.

Table 1 Summary of O-D calibration research approaches

Categories	Approaches	Representative past literature	Approach of this research
Based on model formulation	Least square bi-level type of two-norm formulation	Sherali and Park 2001; Yang et al. 1992; Zhou and Mahmassani 2006, Verbas et al. 2011; Shao et al. 2015; Shen and Wynter 2012;	Adopts one-norm formulation that is more computationally effective and applicable to generic networks
	SPSA formulation	Cipriani et al. 2011; Lu et al. 2015; Tympakianaki et al. 2015;	
	One-norm formulation	Sherali et al. 1997; Hu 2013;	
Based on available data and/or calibration target	Traffic counts	Carey et al. 1981; Cascetta et al. 2013; Zhou and Mahmassani 2007; Bierlaire and Toint 1995;	Field observed data is utilized in the demand calibration process, and calibration result will match with field observed time-dependent speed profile
	Time dependent speed and/or density	Ashok and Ben-Akiva 2002; Nie and Zhang 2008; Qian et al. 2012	
	Other source of data	Iqbal et al. 2014; Parry and Hazelton 2012; Zhou and Mahmassani 2006;	
Based on split rate	Assumes constant proportion	Carey et al. 1981; Cascetta et al. 2013; Bierlaire and Toint 1995;	Split rate is adjusted in the first-stage model. The second-stage model fixes split rate unchanged and adjusts vehicle departure time profile instead.
	Incorporates traffic assignment to determine the O-D proportion	Yang et al. 1992; Ashok and Ben-Akiva 2002; Nie and Zhang 2008;	
	Builds optimization models that does not require explicit link-path incidences	Lu et al. 2013; Frederix et al. 2011; Frederix et al. 2010	
Based on application	O-D calibration with real-time traffic condition	Antoniou 2004; Mahut et al. 2004; Flötteröd et al. 2011	Has the potential to be deployed in an online fashion
	Online and offline deployment	Balakrishna et al. 2007; Balakrishna et al. 2007; Wen et al. 2006;	

III. METHODOLOGY

For convenient reference purposes, the mathematical notations used in this section are presented below first. Some notations from [21] are adopted for the sake of consistency.

Set:
W: Set of O-D pairs that associate with calibration links
L: Set of links to be calibrated
H_d: Set of discretized departure time intervals
H_o: Set of discretized observation intervals
K: Set of modes. 0 for auto and 1 for truck

Index:
w: Index of O-D pairs, $w \in W$
l: Index of links to be calibrated, $l \in L$
t: Index of simulation time intervals, $t = 0, 1, \dots, T$
τ: Index of departure time intervals, $\tau \in H_d$
k: Vehicle mode. $k = 0$ for auto and $k = 1$ for truck
c: Index of calibration iteration, $c = 0$ for initial O-D trips

Locations:
loc: Location of interest, $loc \in \{b, u, o\}$, where b is the bottleneck location, u is the bottleneck upstream location which is the farthest location of the slowly moved vehicle due to shockwave propagation, and o is the origin location where the vehicle departs.

Parameters:
d_w^k: Number of total trips in the planning horizon for O-D zone pair w for mode k in the initial O-D matrix
$d_{l,w}^k$: d_w^k on calibration link l
$d_w^{k,t}$: d_w^k at time interval t
g_l^k: Field observed traffic counts for mode k on link l
g_l : Field observed total traffic counts on link l
q^t : Flow rate at a specific location or link at time t
k^t : Density at a specific location or link at time t
$alpha$: Step size
α^k : Tolerable trip deviation percentage for mode k
β^k : Tolerable total trip deviation percentage for mode k

Variables:
x_w^k: Number of trips estimation for O-D pair w for mode k
$x_w^{k,\tau}$: x_w^k at departure time τ
$d_w^{k,t,c}$: Estimated $d_w^{k,t}$ at iteration c
$p_{l,w}^k$: Link flow proportion for $d_{l,w}^k$ over d_w^k
N_{loc}^t: Cumulative traffic counts at time t at location loc
TT^τ: Travel time between any 2 locations leaving at time τ

A simulation based dynamic traffic assignment (SBDTA)-based two-stage calibration model is built in this research. Figure 1 shows its overall framework, which can be regarded as a sequential two-stage formulation, with the first-stage problem being to estimate O-D trip rates by minimizing link demand deviation with a one-norm formulation approach, aiming to efficiently match the traffic demand on calibration links with the link demand from the field data. The decision variable of first-stage model is x_w^k, i.e. the number of trip estimation for mode k for O-D pair w. Once x_w^k is known, second-stage will further calibrate its temporal profile $x_w^{k,\tau}$ iteratively, i.e. the number of trip estimation at departure time τ for mode k for O-D pair w. The second-stage problem is formulated as a time-dependent, user equilibrium traffic assignment model to adjust the departure time profile, aiming to match the calibrated result with the field observed dynamic traffic condition, i.e. time dependent traffic speed profile.

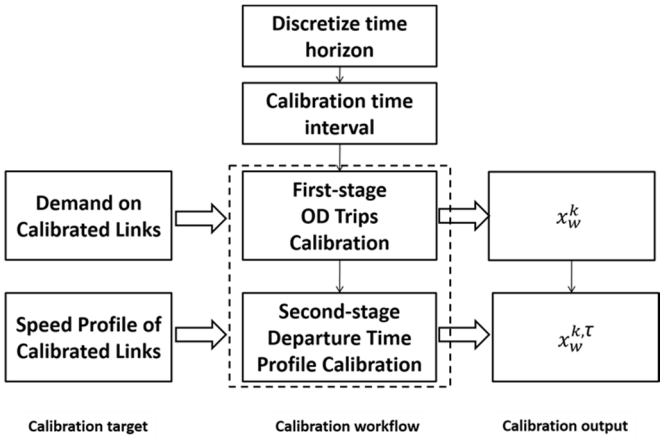


Figure 1: Framework of two-stage calibration model

It's worth noting that the two-stage formulation in this research differs from the classical bi-level programming formulation, that the latter approach is usually formulated with the lower level problem embedded (or nested) within the upper level problem, while the two-stage model in this paper is sequential, with the result from first-stage as the perquisite and directly affects the quality of second-stage calibration. In other words, the second-stage model assumes the total link demand between base scenario and field data are rather close, and the difference in speed profile and congestion pattern can be mainly attributed to the difference in departure time profile. With such dependency designed, the calibrated demand from first-stage will be passed to (and thus, connects first-stage with) second-stage model. Such sequential multi-stage process design may also add values in building a practically efficient approach for agencies, since sometimes due to concerns including data availability, time constraints or budget limit, they may have desire for a phased implementation.

Before we go further to present the overall methodology, the following three main assumptions apply:

- *Assumption 1: Field data exist for road segments to be calibrated over the calibration period*
- *Assumption 2: User Equilibrium principle is able to capture the real world traffic condition*
- *Assumption 3: Evaluation of the proposed calibration model will only be performed on the calibration target links where field data is available.*

A. First-stage methodology

The first-stage problem is formulated as a one-norm formulation to minimize the absolute difference between the estimated and actual link demand. Moreover, unlike other prior formulations that minimize the weighted measures of deviation, it was constrained in the constraint set instead of being specified in the objective function. These constraints

include user-specified tolerable deviations for zone pairs as well as for total trips. The modeling process starts from the algebraic expression of the one-norm linear problem in eq. (1), where $p_{l,w}^k = d_{l,w}^k / d_w^k$ is the link-flow proportion for O-D pair w on link l and is only an approximation to highly dynamic assignment matrix:

$$\text{Minimize } \sum_{k=0}^{|K|} \sum_{l=1}^{|L|} \left| \sum_{w=1}^{|W|} \left(p_{l,w}^k x_w^k\right) - g_l^k \right| \quad (1)$$

Eq. (1) can be reformulated to be eq. (2) plus two constraints (3) - (4) by introducing slack variables h_l^k.

$$\text{Minimize } \sum_{l=1}^{|L|} \sum_{k=0}^{|K|} h_l^k \quad (2)$$

$$\sum_{w=1}^{|W|} \left(p_{l,w}^k x_w^k\right) - g_l^k \le h_l^k \ \forall l \in L, \mathrm{k} \in \mathrm{K} \quad (3)$$

$$-\left[\sum_{w=1}^{|W|} \left(p_{l,w}^k x_w^k\right) - g_l^k\right] \le h_l^k \ \forall l \in L, \mathrm{k} \in \mathrm{K} \quad (4)$$

Next, from equation (3), a new slack variable v_l^k is introduced to yield equations (5).

$$\sum_{w=1}^{|\mathrm{W}|} \left(p_{l,w}^k x_w^k\right) - g_l^k - h_l^k + v_l^k = 0 \quad (5)$$

Substitute h_l^k in equations (3)-(4) with $\sum_{w=1}^{|\mathrm{W}|} \left(p_{l,w}^k x_w^k\right) - g_l^k + v_l^k$ from (5), the final complete model is presented in equations (6)-(13). Equation (6) is the transformed linear objective function. Equation (8) ensures the trip estimation for each O-D pair will not deviate more than α. Equation (9) ensures the total trip estimation will not deviate more than β. Equations (10)-(11) ensure the variables take non-negative values. Equation (12) requires the value of α and β to be within 0 and 1. Equation (13) indicates a TDUETA process to map O-D to the calibration link demand.

$$\text{Minimize } \sum_{l=1}^{|L|} \sum_{k=0}^{|K|} h_l^k \quad (6)$$

Subject to:

$$-2\left[\sum_{w=1}^{|W|} \left(p_{l,w}^k x_w^k\right) - g_l^k\right] - v_l^k \le 0 \ \forall l \in L, \mathrm{k} \in \mathrm{K} \quad (7)$$

$$(1-\alpha^k) d_w^k \le x_w^k \le (1+\alpha^k) d_w^k \ \forall w \in W, \mathrm{k} \in \mathrm{K} \quad (8)$$

$$(1-\beta^k) \sum_{w=1}^{|W|} d_w^k \le \sum_{w=1}^{|W|} x_w^k \le (1+\beta^k) \sum_{w=1}^{W} d_w^k \ \forall \mathrm{k} \in \mathrm{K} \quad (9)$$

$$x_w^k \ge 0 \quad \forall w \in W, \mathrm{k} \in \mathrm{K} \quad (10)$$

$$v_l^k \ge 0 \quad \forall l \text{ in } L, \mathrm{k} \in \mathrm{K} \quad (11)$$

$$1.0 \ge \alpha \ge 0, 1.0 \ge \beta \ge 0 \quad (12)$$

$$G = \varphi(x_w^k, x_w^1, \forall w \in W, \mathrm{k} \in \mathrm{K}) \quad (13)$$

B. Second-stage methodology

We know when congestion happens, volume output will equal to or lower than roadway capacity, and the difference is the number of vehicles in the queue. In this research, we adopt an innovative approach to utilize field data in the calibration process, specifically, start from the demand-capacity-volume relationship at a congested road segment, utilize shockwave theory to capture the differences between demand and volume, and derive an estimate of "arrival" or "demand" information for the bottleneck locations that need calibration. Such demand information, based on the concept of travel time propagation between origin and bottleneck locations, through proper spatial-temporal mapping, enables to infer the traffic demand departure time pattern at various origins, i.e. $x_w^{k,\tau}$.

Input-output N-Curve is used to construct the demand curve, which is the integral of $x_w^{k,\tau}$ with regard to τ. Figure 2 illustrates the relationship between N-curves at location b, u and o, where X-axis stands for time dimension, and Y-axis is the cumulative number of vehicles passes a certain location.

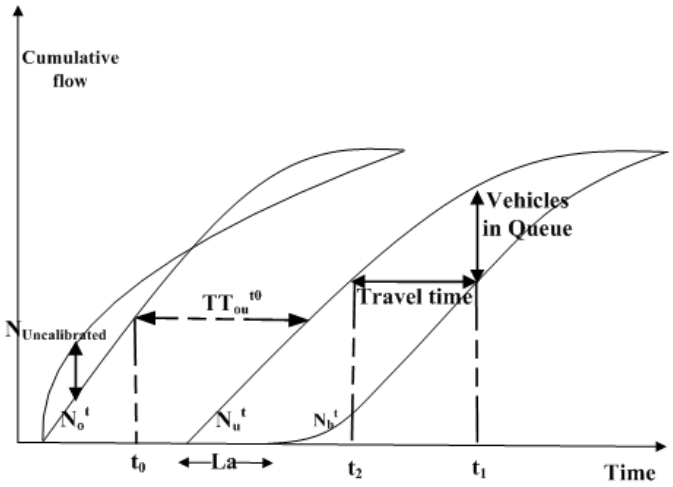


Figure 2: Cumulative number of vehicles at 3 locations

A few observations from this diagram:

1. It's intuitive to tell N_b^t is the cumulative traffic counts (volume output) that can be observed at the bottleneck location, i.e. N_b^t can be easily calculated from the data, but it is the unobservable N_o^t that represents the real demand and needs to be calibrated
2. At any given time t_1, the difference between two N-Curves at bottleneck location is the number of vehicles in queue. In other words, N_u^t can be inferred based on N_b^t
3. For any given cumulative vehicle count N, the horizontal difference between N_o^t and N_u^t is the time needed for the vehicles to travel from location o to location u. In other words, N_o^t can be inferred based on N_u^t
4. Once N_o^t (i.e. calibrated N-curve) is known, we should correct initial O-D demand ($N_{uncalibrated}$) and rerun DTA model until system converges.

C. Second-stage calibration procedure

1) Step A: Estimate N-curve at bottleneck location b

By observation 1, we have eq. (14) where q_b^t is the volume data observed at bottleneck location b.

$$N_b^t = \int_0^t q_b^t dt \quad (14)$$

2) Step B: Estimate N-curve at upstream location u

By observation 2, we have eq. (15) where time-dependent travel time between location u and b is denoted as $TT_{u,b}^{\tau}$, $\bar{k}_a^t$ is the average density on segment a between u and b, L_a is its length and LN_a is the number of lanes.

$$N_u^t = N_b^{t+TT_{u,b}^{\tau}} + \bar{k}_a^t L_a * LN_a \quad (15)$$

As traffic density in the slow-moving traffic between locations u and o does not fluctuate much, $\bar{k}_a^t$ can be approximated by the density measured at location b,

$$\bar{k}_a^t \approx k_b^t \quad (16)$$

Next, Eq. (17) can be used to estimate the only remaining unknown variable L_a , where $\omega_{u,b}(t)$ is the shockwave propagation speed at time t, and the time of interest is $[0, H_o]$.

$$L_a = max \int_0^{H_o} \omega_{u,b}(t) dt = max \int_0^{H_o} \frac{q_u^t - q_b^t}{k_u^t - k_b^t} dt \quad (17)$$

It's worth noting that Eq. (17) is a simplified formulation to illustrate the idea. In reality, there may exist multiple shockwaves, including those backward and forward shockwaves, and based on the dynamic traffic conditions, the speeds of shockwaves are time-dependent. Classic shockwave propagation process could be referred to [30], and we know maximum queue length happens when queuing shockwave and discharging shockwave meet. Considering the paper length limitation, these details are not repeatedly cited.

3) Step C: Estimate N-curve at origin location o

By observation 3, we have eq. (18), where time-dependent travel time between location o and u is denoted as $TT_{o,u}^{\tau}$

$$N_o^t = N_u^{t+TT_{o,u}^{\tau}} \quad (18)$$

Combining Eq. (14)-(18), we can derive Eq.(19) as:

$$N_o^t = N_u^{t+TT_{o,u}^{\tau}} = N_b^{t+TT_{o,u}^{\tau}+TT_{u,b}^{\tau}} + k_b^{t+TT_{o,u}^{\tau}} LN_a * max \int_0^{H_o} \frac{q_u^t - q_b^t}{k_u^t - k_b^t} dt \quad (19)$$

4) Step D: Adjust O-D temporal profile in DTA model

From observation 4, by comparing the difference between calibration result N_o^t and initial demand $N_{uncalibrated}^t$, discrepancy can be observed and we need to adjust the vehicle departure time profile per the equation below.

$$N_{new}^t = N_{uncalibrated}^t + (N_o^t - N_{uncalibrated}^t) * alpha \quad (20)$$

5) Step E: Run DTA simulation model with DUE

With the new calibration result, DTA model will run to reach dynamic user equilibrium. Link statistics including speed, density and volume will be updated accordingly.

6) Step F: Check if calibrated speed profile is satisfactory

If the time-dependent speed profile deviation is lower than pre-defined threshold, go to step G, otherwise go to step A.

7) Step G: Obtain time-dependent O-D demand matrices

Once a satisfactory N_o^t has been calibrated, the final time-dependent demand matrices can be calculated as below. Note at origin location, τ equals to t.

$$x_w^{k,\tau} = dN_0^t/dt \quad (21)$$

IV. CASE STUDY

A. Scenario setup

A case study based on Tucson network was conducted as an experimental proof of concept to demonstrate the proposed model's performance. A DTA simulation model DynusT was adopted. The network includes 80 traffic analysis zones, 830 links and 395 nodes. 16 links were selected as the first-stage calibration objects, and second-stage picks one freeway link and one arterial link as calibration targets. The total calibration time is 4 hours, with 15 minutes as time interval.

Initial demand (noted as the un-calibrated case) is purposely constructed to be significantly different compared with ground truth (GT) scenario. Specifically, structure-wise this paper purposely built an initial solution with departure time profile distorted systematically, with peak hours pushed forward so that at bottleneck locations the traffic congestion pattern appears earlier. Amount-wise, the total number of vehicles in the initial solution is also set to deviate from ground truth data by certain extent. Initial scenario can be found as black square in Figure 3 and red dots line in Figure 4.

B. First-stage results analysis

First-stage algorithm is coded in MATLAB and Python and converges at the 9th iteration. After calibration, the volume comparison result is shown in Figure 3. X-axis indicates the traffic demand from the sensor, i.e. the GT data. Y-axis is the Link traffic demand before and after first-stage calibration.

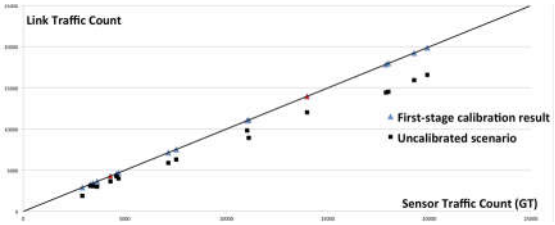


Figure 3: First-stage calibration result

The first-stage calibration goal is to move the black squares (i.e. un-calibrated links) as close to 45-degree line as possible. Based on Figure 3, it's safe to conclude after first-stage calibration (triangle dots), the demand on the calibration links has been matched quite well compared with GT scenario.

C. *Second-stage results analysis*

Second-stage is coded in Java. The calibration model converges at the 96th iteration. The time-dependent speed profile comparison result is shown in Figure 4 (a) and (b).

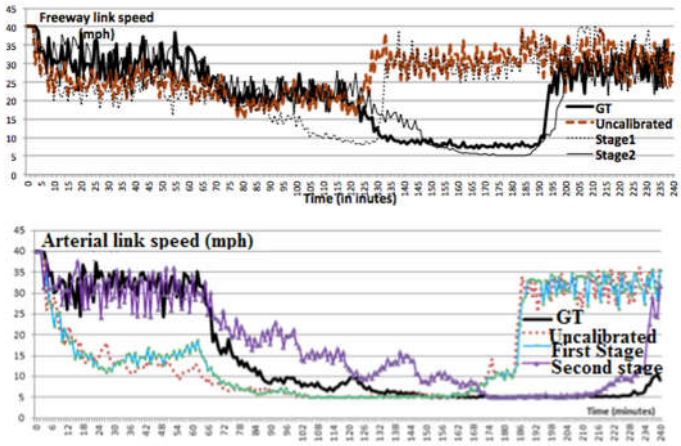
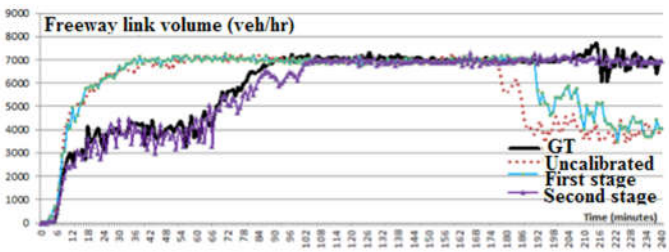


Figure 4: Speed profiles on a) freeway link, b) arterial link

From the above figure, one can easily tell the significant speed profile improvements: before calibration, the speed drop on the freeway link in the Un-calibrated case (red dot line) happened around the 20th minute, and even after we managed to match the total traffic demand on this link after first-stage calibration (most prior studies with traffic counts as the only calibration target will stop at this step), the speed profile is still very different - the speed drop (blue line) is about 1 hour before the GT case (black line). Such discrepancy can also be observed on the arterial link, where heavy congestion happened from the 1st hour although in the real case the traffic is light at that time. However, after second-stage calibration, the calibrated speed temporal distribution is getting very close to the GT case for both freeway and arterial links, evidenced by the purple lines staying very close to the black lines. This promising result is achieved without the need of utilizing additional data, other than commonly used volume and density information at bottleneck location.

We can also visually inspect the time-varying volume profile between the black curve (GT case) and the purple curve (second-stage result). Figure 5 (a) and (b) also confirm, the second-stage calibration result is able to match with GT scenario very well. These figures also show the discrepancies of the result from first-stage, i.e. even if we managed to match the link demand as closely as possible, the difference in temporal statistics for the link can still be significant, which is exactly the issue the second-stage calibration tries to address.



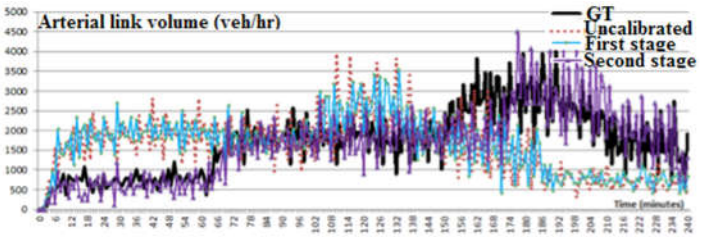


Figure 5: Volume profile on a) freeway link b) arterial link

Vehicle cumulative departure N-curve is also examined. It can be found that the blue N-curve (first-stage result) continues to differ significantly from GT scenario (black). However, one can tell the purple curve (second-stage result) becomes much closer to the black curve, which reiterates that the proposed two-stage calibration model, despite the large differences between GT scenario and initial solution, is able to converge to a good solution that matches well with GT.

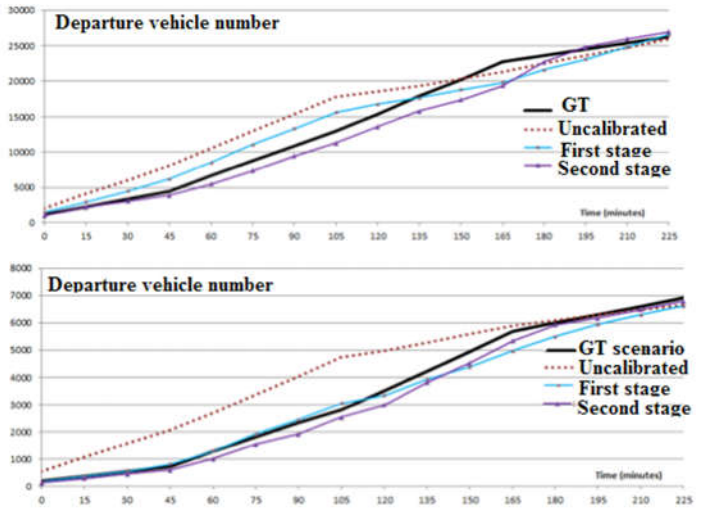


Figure 6: Vehicle cumulative departure profile on a) freeway link b) arterial link

Mean Absolute Error (MAE) with regard to speed profile is also analyzed to show the convergence property of the algorithm. Figure 7 shows that MAE values for both freeway calibration link and arterial calibration link drop rapidly in the first few iterations and then quickly becomes stable, demonstrating that the proposed two-stage calibration model is able to converge very quickly.

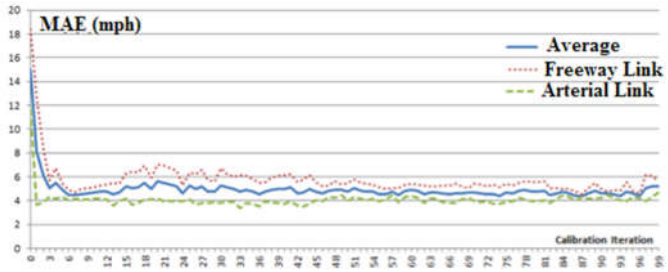


Figure 7: MAE with regard to speed profile

V. DISCUSSIONS AND CONCLUDING REMARKS

The research effort in developing a two-stage model to calibrate the time-dependent, dynamic O-D demand for a congested network is documented in this paper. Compared with the commonly seen least-square type of two-norm formulations, the one-norm formulation proposed is more computationally effective due to its linear model structure. The case study also demonstrates the effectiveness of the proposed methodology in utilizing commonly available volume and density data at bottleneck location in modeling process, and its advantages over prior studies that either stop at first-stage or have to rely on other supplemental data.

One freeway link and one arterial link were selected as second-stage calibration targets. While we are glad to see the case study results is satisfactory, if to extend second-stage model with more calibration links, the interactions between multiple calibrated links could be complicated. How to model such interactions in greater details and its influence in the calibration result is certainly another interesting topic to study as part of future research direction.

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